

Cross-Domain Compositional Schema

Future-Sufficient Interfaces, Load-Bearing Relations, and Preserve-or-Reopen Reuse

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Publication boundary

This paper presents a bounded architectural, translational, and formalised interface result. It does not claim a new law of nature, a universal ontology, a new Bell inequality, a new Tsirelson bound, a new quantum mechanism, or superiority over domain-native methods. Version 0.4 reopens the tetrahedral Bell branch left deliberately unresolved in v0.3 and carries it to an exact mathematical fixed point. The result is not a tetrahedral ontology: it distinguishes an abstract four-context indexing simplex from the established three-dimensional CHSH facets of the four-dimensional local correlator polytope, then defines an exact facet-adapted tetrahedral-plus-normal coordinate chart. The chart is a reparameterisation of known Bell objects. Its natural four-dimensional hypervolume reduces exactly to the CHSH excess and therefore supplies no independent new physical invariant. Version 0.4 carries DOI 10.5281/zenodo.22166468 within the same Zenodo version lineage as v0.3 (10.5281/zenodo.22166005).

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Abstract

This paper formalises a typed cross-domain compositional schema for deciding when a compressed or closed object may be reused inside a wider composition without silently losing a relation that later becomes decision-bearing. Relative to a declared future-operation family F , a higher-level interface may compress lower detail only if it preserves the declared evaluations or carries a reliable trigger and descent address that restores sufficient lower structure before an omitted distinction is used. The formal object is $X = (N, R, \tau, \alpha)$; evaluation is three-part, $\text{Eval}_k(X) = (s_k, v_k, \rho_k)$; exact relation ablation is partial and protocol-relative; and load-bearingness is defined by change in a declared evaluation rather than by visual prominence or post-hoc interpretation. Version 0.4 additionally formalises **operator-to-target attachment**: a modifier or sign is not a complete operational object when its receiver, layer, scope, or address can change the result.

The existing cross-domain tests - correlated metrology, software dependency composition, and a formal-proof refusal control - are retained. The Bell/CHSH section is then taken beyond the v0.3 Phase-0 dimension check. Two different tetrahedral objects are separated. First, the four Bell contexts $\mathcal{C} = \{00, 01, 10, 11\}$ form an abstract indexing simplex whose Boolean face lattice records which context-labelled quantities are exposed. A CHSH sign section $\sigma : \mathcal{C} \rightarrow \{-1, +1\}$ with odd parity attaches coefficients to exact contexts and selects a Bell functional $S_\sigma(E) = c_\sigma^\top E$. Erasing that attachment while retaining only the sign multiset loses the facet selector. Second, Bell-native geometry supplies a genuine tetrahedron: each CHSH facet of the local correlator polytope in $\mathbb{R}^4$ is a three-dimensional simplex with four deterministic local vertices.

For the standard orientation $c = (1, 1, 1, -1)^\top$, let V contain the four local vertices saturating $c^\top E = 2$. Define the signed CHSH excess $\nu = c^\top E - 2$, orthogonal projection $\Pi_c(E) = E - (\nu/4)c$, and barycentric coordinate $\lambda = V^{-1}\Pi_c(E)$. Then

$$E = V\lambda + \frac{\nu}{4}c, \quad \mathbf{1}^\top \lambda = 1.$$

This **Facet-adapted Tetrahedral Bell Chart (TBC)** resolves the v0.3 four-versus-three degree-of-freedom obstruction without contradicting it: the tetrahedral base carries three independent coordinates and the transverse Bell coordinate ν carries the fourth. On the local CHSH facet, $\nu = 0$ and λ is an ordinary non-negative barycentric point. Away from the facet the chart remains exactly invertible, although λ need not remain inside the simplex.

The construction is pressure-tested rather than promoted by resemblance. Along the outward normal through the facet centroid, the local boundary occurs at $\nu = 0$, the symmetric Tsirelson point at $\nu = 2(\sqrt{2} - 1)$, and the PR/no-signalling algebraic point at $\nu = 2$. Pulling the Tsirelson-Landau-Masanes correlator criterion into (λ, ν) coordinates changes no native quantum boundary. Likewise, extending back to full binary behaviours requires the marginal fibre of the projection $P \mapsto E$; identical TBC coordinates can coexist with different signalling marginals. Finally, the natural four-dimensional simplex volume formed by a CHSH facet and an arbitrary point E satisfies

$$\text{Vol}_4(\text{conv}\{v_1, v_2, v_3, v_4, E\}) = \frac{|S(E) - 2|}{3},$$

so the most immediate geometric “new invariant” collapses exactly to the already-known CHSH excess. The TBC is therefore a precise and potentially useful representation, not a new Bell law. No new Bell inequality, independent Tsirelson derivation, quantum mechanism, or measurable physical residual survives the present native-owner subtraction. The paper closes the tetrahedral branch at the level of exact semantics and geometry while keeping any future new-physics branch conditional on an additional non-reparameterisation law with a prospective observable discriminator.

Keywords: cross-domain composition; future sufficiency; load-bearing relation; exact relation ablation; semantic attachment; interface contract; restorative descent; recursive reuse; parent completion; Bell; CHSH; tetrahedral Bell chart; local correlation polytope; no-signalling; baseline parity.

1. Scope, claim, and non-claims

The paper asks a narrow question:

When may a lower object be compressed into a higher-level interface and then reused in a wider composition without losing a relation that later becomes decision-bearing?

The proposed answer is not a new universal data structure. It is a **cross-domain compositional schema**: a common audit contract that makes the same kinds of closure questions explicit across heterogeneous domains while permitting domain-specific fields to remain distinct or absent.

The current claim is architectural and translational. The schema is useful only if it can do at least three things simultaneously:

1. preserve a common functional contract across domains;
2. refuse false analogy by leaving genuinely absent fields absent; and
3. fail cleanly when a load-bearing relation is removed or when a target mapping requires semantic substitution.

The paper does **not** claim a new Bell inequality, a new Tsirelson bound, or a new quantum mechanism. It does **not** claim that physical reality is literally a lattice, simplex, tetrahedron, fibre bundle, or higher-dimensional geometric object. It does not claim that three inputs are privileged. It does not claim that an embedded observer contributes information that is metaphysically irreducible to a sufficiently complete and current representation. It does not claim that representation of authority transfers authority. It does not claim that the present schema outperforms strong domain-native formalisms.

The originating geometric language remains useful as a historical scaffold: a lower relation can close into an addressable object, and that object can later participate in another relation. The present paper treats that role change as a typed interface/composition problem rather than evidence of literal dimensional lift.

Operationally, the target failure is **interface loss during recursive reuse**. A closure that was locally sufficient can cease to be future-sufficient when a wider composition makes a previously hidden dependency, unresolved remainder, provenance relation, or authority/admissibility relation decision-bearing. The schema is intended to make that transition explicit early enough either to preserve the relation or to reopen only the affected lower structure, rather than carrying all lower detail indefinitely or discovering the omission only after upper reuse.

2. From geometric scaffold to compositional object

The originating worked cell used three lower roles - an inside/participating description, an outside/contextual description, and a coupling/interface description - with a fourth point representing their closure. Pressure-testing removed two assumptions that had initially been visually attractive:

- **privileged arity three**: the same closure discipline must work for independently justified arities other than three;
- **literal dimensional lift**: a role-changing abstraction or new addressable interface does not by itself prove an increase in mathematical or physical dimension.

The surviving object is cardinality-neutral. Let a lower relational object be

$$X = \{N_i, R_j\}$$

where the N_i are typed lower objects and the R_j are the relations among them that remain relevant to a declared future-use family. A compression or closure operator produces an interface

$$Y = C_F(X)$$

relative to a future-operation family F .

The central question is no longer whether Y sits geometrically “above” X . It is whether Y remains a lawful stand-in for X when reused.

This shift also changes the interpretation of hierarchy. A closed object can be a **whole downward** relative to what generated it and a **part upward** relative to a wider composition. Whole/part status is therefore relational and operation-relative rather than a permanent ontological label.

3. The current cross-domain compositional schema

Before a translation test, the common schema is frozen. A target domain may instantiate the following fields when they are operationally real:

1. **Object identity and scope** - what exact object is being closed or reused?
2. **Declared future-operation family** F - for which queries, actions, or compositions is sufficiency being asserted?
3. **Typed dependencies or shared-source relations** - what lower relations can make apparently independent objects coupled at the next level?
4. **Unresolved remainder or uncertainty** - what decision-bearing uncertainty, incompatibility, contradiction, or open state remains, if such a field exists in the domain?
5. **Provenance and descent address** - which lower objects or relations generated the interface, and where can they be reopened?
6. **Admissibility, control, or authority state** - only where a real action-validity relation exists in that domain.
7. **Preserve-or-reopen rule** - what must remain exposed, and what event forces descent?
8. **Recursive reuse contract** - under what conditions may the closed object become an input to a wider composition?

Allowed structural decisions are:

- **PRESERVE** - current interface is sufficient for the declared use.
- **REJECT** - proposed composition is incompatible or invalid as stated.
- **REFINE** - interface must expose additional typed state or relation.
- **REOPEN** - descend to specified lower structure before reuse.
- **WITHHOLD** - information may be sufficient, but action is not currently authorised/admissible.

3.1 Typed partial instantiation

The schema is **partially instantiable**, not universal by forced symmetry. A domain need not contain every field. If a field is not constitutive of the target operation, it remains absent. This is different from deleting a field that was constitutive and then hiding the loss inside a higher-level label.

Optionality is therefore typed rather than discretionary:

Absence is lawful only when the target domain does not instantiate the relation. Omission is unlawful when the relation existed and remains decision-bearing for the declared future operation.

This distinction is the main guard against cross-domain analogy degenerating into vocabulary matching.

3.2 Typed relational object and declared operations

For the formalised contract, write the lower object as

$$X = (N, R, \tau, \alpha).$$

Here $N = \{N_i\}$ is a finite or explicitly bounded family of constituent objects, $R = \{R_j\}$ is a family of typed relations, τ records the type, address, scope, and source/receiver assignment needed to distinguish those objects and relations, and α carries only auxiliary coordinates that are operationally real in the domain. Examples include provenance, admissibility state, authority or delegation state, uncertainty, and restoration address. The tuple is **partially instantiated**: an auxiliary coordinate that is not constitutive of the target operation is absent rather than replaced by an analogy.

Let the declared future-operation family be

$$F = \{f_k\}_{k \in K},$$

where each f_k has an output or decision space D_k and a declared output equivalence or tolerance relation $\approx_k$. The meaning of sufficiency is therefore not global. It is indexed by the operation family against which the interface will be used.

3.3 Evaluation semantics and restoration signature

A binary “defined/not-defined” flag is insufficient for the present architecture because two objects may both require reopening while requiring different lower dependency cones. The evaluation object is therefore

$$\text{Eval}_k(X) = (s_k(X), v_k(X), \rho_k(X)).$$

The three coordinates have distinct jobs:

1. $s_k(X)$ is a typed evaluation state. A domain may use a declared finite set such as **DEFINED**, **INADMISSIBLE**, **RESTORE-REQUIRED**, and **UNDEFINED**, provided the states have operational semantics fixed before the test.
2. $v_k(X) \in D_k \cup \{\perp\}$ is the value or decision result where one is defined. Values are compared only when the evaluation states make that comparison meaningful.
3. $\rho_k(X)$ is the **restoration signature**. It is null when no reopening is required. When reopening is required, it records the family, or an equivalence class, of inclusion-minimal valid restorative descent sets. If a cost preorder was fixed prospectively, it may instead record the cost-minimal family. Multiplicity is preserved; uniqueness is not assumed.

Two evaluations are equivalent, written

$$\text{Eval}_k(X) \simeq_k \text{Eval}_k(X'),$$

only when their typed states agree, their defined values agree within $\approx_k$ where appropriate, and their restoration signatures are equivalent whenever restoration is required. This prevents a higher-level label such as “needs review” from hiding a change in the exact lower structure that must be restored.

3.4 Exact relation ablation and load-bearingness

For a relation R_j , let $\text{Abl}_j(X)$ denote a **lawful exact relation ablation**: remove or neutralise only R_j while preserving the endpoints, object identities, and unrelated comparison coordinates as far as the domain permits. This operator is intentionally partial. Some domains do not admit a meaningful ceteris-paribus relation removal. If no lawful ablation exists, or if multiple equally lawful ablation protocols yield incompatible verdicts and no protocol was selected prospectively, the result remains undetermined rather than being forced.

Define the three-valued load-bearing predicate

$$LB_F(R_j; X) \in \{0, 1, ?\}$$

by

$$LB_F(R_j; X) = 1 \iff \exists f_k \in F : \text{Eval}_k(X) \not\simeq_k \text{Eval}_k(\text{Abl}_j(X)),$$

whenever the ablation and comparison are well-defined. If every declared evaluation is equivalent after a lawful ablation, $LB_F = 0$. If the required ablation or comparison is not well-defined, $LB_F = ?$.

This definition is operational rather than ontological. It does not say that every relation with $LB_F = 1$ is a fundamental causal mechanism. A representation edit may reveal decision-bearing dependence without constituting a physically real intervention. Causal language requires an independent causal model or intervention appropriate to the domain.

3.5 Representation refusal

The formal object has a refusal property. A translation is invalid when a source field can be carried only by changing its functional meaning, when an exact-ablation protocol cannot be stated at the claimed address, or when a genuinely absent target relation is manufactured merely to make the schema look symmetric. The allowed result in those cases is **REJECT** or **ABSENT**, not a rescued analogy.

3.6 Operator-to-target attachment as part of the semantic address

The same address discipline applies to operators and modifiers. Let q be an operator token whose surface identity does not by itself determine what relation it acts on. Define its typed attachment as

$$\beta(q) = (r, \varrho, \ell, s, \xi),$$

where r is the target relation or operand, ϱ is the receiver or role when that changes the operation, ℓ is the layer/domain/address, s is the operator scope, and ξ is an optional state or scale coordinate only when it is decision-bearing. The addressed semantic object is

$$Q = (q, \beta(q)).$$

Let $\pi_{-\beta}$ erase only the attachment, so $\pi_{-\beta}(Q) = q$. For a declared operation family F , define operational equivalence $Q_1 \equiv_F Q_2$ when substituting one for the other leaves every relevant Eval_k equivalent. The attachment is load-bearing whenever

$$Q_1 \not\equiv_F Q_2, \quad \pi_{-\beta}(Q_1) = \pi_{-\beta}(Q_2).$$

Thus the token can remain unchanged while the operational object changes because its receiver, relation, layer, or scope changed. The rule is selective: if syntax, typing, or context fixes the attachment uniquely and erasure cannot collapse distinct live readings, the edge may remain implicit rather than being turned into tagging ceremony.

Where a typed transport T is defined, semantic preservation requires the attachment to travel with the operator:

$$T(q \triangleright r) = T(q) \triangleright T(r).$$

Transporting q while silently dropping or changing the receiver is not semantic preservation. This is a semantic/address custody rule, not a claim that language or signs exert a new physical force. Section 10 gives a mathematically clean Bell instantiation: CHSH signs attached to exact measurement contexts select different facet functionals even though the visible sign inventory can be the same.

4. Future-sufficient interfaces and restorative descent

A higher-level interface need not duplicate every lower detail. It needs to be **future-sufficient** for a declared operation family.

Let X be the fuller lower object, $Y = C_F(X)$ its compressed interface, and f an operation in F . The interface is admissible when either:

1. using Y preserves the f -relevant result or constraint within a declared tolerance; or
2. Y carries a reliable trigger and address that restores the omitted decision-bearing structure before f is allowed to depend on it.

A compact conceptual statement is:

$$\text{Admissible}_F(Y) = \text{Preserve}_F(X, Y) \vee \text{Guaranteed-Reopen}_F(Y \rightarrow X)$$

This is not presented as a novel theorem. It is an interface discipline that combines target-relative sufficiency with explicit restorative descent.

The descent obligation matters because future use can widen. A representation sufficient for one task can become insufficient for another. A closure is therefore not globally “complete” merely because it was sufficient at the time it was created.

4.1 Minimal descent

Reopening everything defeats compression. The stronger requirement is to identify the **minimal restorative descent set**: the smallest lower dependency cone whose recovery is sufficient to restore validity for the widened operation.

The ideal behaviour is:

- local closure $\rightarrow$ wider composition exposes missing relation
- $\rightarrow$ identify affected dependency cone
- $\rightarrow$ reopen only that cone
- $\rightarrow$ re-evaluate the upper closure

This is a practical rather than metaphysical requirement. It can be compared against dependency graphs, provenance systems, package resolvers, proof assistants, and other domain-native mechanisms.

4.2 Formal future-sufficiency contract

Let $Y = C_F(X)$ be a closure of X . For each declared operation, define direct preservation by

$$\text{Preserve}_k(X, Y) = 1 \iff \text{Eval}_k(X) \simeq_k \text{Eval}_k(Y).$$

Then define

$$FS_F(Y \mid X) = \bigwedge_{f_k \in F} [\text{Preserve}_k(X, Y) \vee \text{RestoreBeforeUse}_k(Y \rightarrow X'_k)].$$

The preservation branch therefore requires evaluation-equivalence at the declared target. The restoration branch requires, before the omitted distinction becomes consequential, a deterministic or otherwise prospectively reliable trigger, a descent address, and recovery of a sufficient lower object X'_k whose evaluation is equivalent to that of X for the declared operation.

This is a contract definition, not a new information-theoretic theorem. Its added burden is explicit sequencing: an interface may be deliberately incomplete only when the omitted structure is either irrelevant to the declared use or guaranteed to be restored before the widened use depends on it.

4.3 Recursive reuse and promotion

Let

$$X_0 \xrightarrow{C_{F_0}} Y_0, \quad Y_0 \in N(X_1), \quad X_1 \xrightarrow{C_{F_1}} Y_1.$$

A relation can satisfy $LB_{F_0}(R; X_0) = 0$ while satisfying $LB_{F_1}(R; X_1) = 1$ after Y_0 is reused inside the wider parent. This is **recursive promotion**: a relation that was not decision-bearing for the local operation family becomes decision-bearing for the widened family. The mechanism is not mysterious; it is the ordinary consequence of changing the question while reusing a compressed object. The formal discipline is to make the change of operation family explicit rather than silently carrying the earlier closure certificate upward.

Proposition 1 - Operation-relative reopening. Let $Y_0 = C_{F_0}(X_0)$ omit an addressed relation R_j . If R_j is non-load-bearing for F_0 , but a widened composition admits an operation $f \in F_1$ for which the corresponding relation becomes load-bearing, then future sufficiency for F_0 alone does not establish future sufficiency for F_1 . Before f depends on that distinction, the widened interface must either expose the required relation or satisfy the declared restoration condition.

Justification. Exact ablation of the relation changes an F_1 -relevant evaluation by assumption. The earlier F_0 closure certificate therefore cannot supply the missing F_1 equivalence. The widened family must satisfy the preserve-or-restore contract on its own terms. $\square$

This is an elementary consequence of the definitions, but it isolates the precise promotion error: **local closure is not a licence for unrestricted future reuse**.

4.4 Parent completion and obstruction

Suppose local constituents satisfy local admissibility predicates A_i . For a parent operation family F_P , let R_{F_P} denote the cross-relations that are relevant to the parent question, and let Φ_j be the domain-native constraint induced by relation R_j . Define

$$\text{Comp}_{F_P}(X) = 1 \iff \exists Z \in \Omega_P : \left[\bigwedge_i A_i(Z \mid N_i) \right] \wedge \left[\bigwedge_{R_j \in R_{F_P}} \Phi_j(Z) \right].$$

Local PASS therefore does not imply parent completion. If $\text{Comp}_{F_P}(X) = 0$, an obstruction set is any active subset of local or cross constraints that is jointly unsatisfiable. Inclusion-minimal obstruction sets need not be cardinality-minimal and need not be unique.

There is no claim that this existential predicate is a new satisfiability formalism. Constraint-satisfaction problems, minimal-unsatisfiable subsets, model-based diagnosis, and sheaf/local-to-global consistency already own the native mathematics of this address [14,20,21]. Parent completion is used here as a typed wrapper that exposes **when** a locally closed object fails to compose and **where** restorative descent must be addressed.

4.5 Restorative descent

Let $\mathcal{D}(Y, F)$ be the family of lawful descent sets that can be reopened from interface Y for operation family F . A descent set Δ is valid if

$$FS_F(\text{Reopen}(Y, \Delta) \mid X) = 1.$$

If a defensible cost function or preorder was fixed before inspection, one may select

$$\Delta_F^*(Y) \in \arg \min_{\Delta \in \mathcal{D}(Y, F)} \text{Cost}(\Delta)$$

subject to validity. Where no defensible scalar cost exists, the correct object is the **family of inclusion-minimal valid descent sets**. The paper therefore does not assume uniqueness or a universal metric of “smallest.”

Proposition 2 - Target-scoped reopening. Suppose Δ is a valid restorative descent set for a declared target operation and every changed evaluation or active obstruction that must be repaired for that target lies within the dependency cone addressed by Δ . Then components outside Δ are not required by the restorative proof for that target, unless reopening Δ changes the comparison object, activates an additional load-bearing relation, or exposes another active obstruction relevant to the same target.

Justification. Validity already requires restoration of future sufficiency for the declared target. Under the stated dependency assumption, information outside Δ is not used by that proof. The exceptions prevent the scope statement from being mistaken for a claim that reopening is causally isolated or globally sufficient. $\square$

This is a **scope condition**, not an algorithmic speed guarantee. Finding a valid or minimal descent set can remain computationally difficult, and a different target may require a different cone.

4.6 Elementary structural lemmas

The following results are direct consequences of the definitions. They are useful because they make the contract checkable, but they are not presented as deep new theorems.

Lemma 1 - Typed readdressing invariance. Let ϕ be a bijective readdressing that preserves relation identity/type/scope, preserves the declared evaluations up to their target equivalences, and commutes with exact ablation:

$$\phi(\text{Abl}_j(X)) = \text{Abl}'_{\phi(j)}(\phi(X)).$$

Then, for determinate cases,

$$LB_F(R_j; X) = LB_{F'}(\phi(R_j); \phi(X)).$$

Proof. A witness evaluation inequivalence exists before readdressing if and only if the corresponding inequivalence exists after readdressing, by evaluation preservation and the commuting-ablation assumption. $\square$

Lemma 2 - Operation-family monotonicity of load-bearingness. If $F \subseteq G$ and the common evaluation and ablation semantics are fixed, then

$$LB_F(R; X) = 1 \implies LB_G(R; X) = 1.$$

Proof. The witnessing operation in F is also an operation in G . The converse can fail because G can contain a new operation for which the relation becomes decision-bearing. $\square$

Lemma 3 - Restriction monotonicity of future sufficiency. If $F \subseteq G$, then

$$FS_G(Y \mid X) = 1 \implies FS_F(Y \mid X) = 1.$$

Proof. The conjunction required for G includes every conjunct required for F . The converse can fail when the widened family asks a question that the compressed interface cannot answer or restore in time. $\square$

Lemma 4 - Parent-completion monotonicity under constraint enlargement. If the active local/cross-constraint family for F is a subset of that for G , then

$$\text{Comp}_G(X) = 1 \implies \text{Comp}_F(X) = 1.$$

Proof. Any parent witness satisfying the larger active constraint set satisfies the subset. $\square$

Lemma 5 - Minimal-descent existence in the finite case. If the family of valid restorative descent sets is finite and non-empty, it contains at least one inclusion-minimal member. If, additionally, a real-valued cost is defined on that finite family, at least one cost minimiser exists. Neither statement implies uniqueness.

Proof. Every finite non-empty partially ordered set has a minimal element; every real-valued function on a finite non-empty set attains a minimum. $\square$

5. Search, conditioning, admissibility, authority, and action

A second pressure-test concerned observer inclusion and local authority. The strong claim that an embedded observer necessarily contributes irreducible information did not survive. A sufficiently complete, current representation can in principle encode descriptive local state, history, goals, constraints, and other variables relevant to admissibility.

What survives is a typed separation of operations:

SEARCH $\rightarrow$ CONDITIONING $\rightarrow$ ADMISSIBILITY $\rightarrow$ AUTHORITY $\rightarrow$ ACTION

- **SEARCH** produces candidate pathways or descriptions.
- **CONDITIONING** supplies the state, history, target, and constraints that make those pathways locally interpretable.
- **ADMISSIBILITY** determines which candidates remain valid for the declared use.
- **AUTHORITY** determines who or what may lawfully select, withhold, or delegate an action, where such a relation exists.
- **ACTION** executes the selected transition.

Search height can widen the first operation without automatically supplying the later four. More importantly, representing or predicting an authority-bearing relation does not itself enact that authority. A valid delegation is a separate, scoped, time-sensitive transition with provenance.

This authority field is not mandatory in every domain. The formal-proof negative control below deliberately tests a domain where theorem validity does not instantiate it.

6. Component precedent versus integrated-schema utility

Many ingredients of the present schema have strong precedents:

- sufficient statistics and task-relative compression formalise retention of information relevant to an inference target [1,2];
- factor graphs and dependency graphs make local relations and shared dependencies explicit [3];
- sheaf-based information fusion addresses heterogeneous local information, overlap consistency, and faithful local-to-global integration [4];
- compositional/open-system formalisms model interfaces that hide internal structure while retaining external behaviour needed for further composition [5];
- metrology already requires explicit uncertainty, covariance, calibration, and traceability [6];
- software supply-chain systems preserve provenance and dependency relationships across build/release stages [7];
- access-control formalisms separate possession of information from valid permission to act [8];
- proof assistants make theorem validity dependent on an explicit formal environment and trusted checking base rather than editorial authority [9].

These precedents sharply constrain novelty claims. The current schema does not become a new mathematical formalism merely by combining familiar mechanisms.

However, **component precedent and integrated-schema utility are different questions**. Showing that each part has a formal home does not, by itself, establish that one frozen role/relation contract already exists as a practical cross-domain audit object or that such a contract is useless. The translational burden is therefore independent:

Can the same compositional object be stated once, mapped into operationally unlike domains without changing the load-bearing function of its relations, and allowed to fail when a target lacks a counterpart?

Formal novelty and architectural/translational usefulness are separate axes.

7. Translation-fidelity protocol

A cross-domain translation is treated as nontrivial only when the source contract is frozen before the target outcome is inspected.

Here, **the same compositional object** means preservation of the frozen role/relation contract and its decision-bearing functions under translation. It does not mean that source and target are the same historical token or entity, nor that they instantiate identical local semantics. Token identity and provenance remain separately tracked and must be independently established where they matter.

For a source domain S and target domain T :

1. freeze the common schema and allowed decision classes;
2. define the source object and its failure/ablation prospectively;
3. construct target correspondences from the target domain's own operational semantics;
4. leave absent fields absent rather than manufacturing analogues;
5. pre-register a perturbation or ablation before evaluating its outcome;
6. ask whether both domains produce the same structural decision class for the same functional reason;
7. compare the schema decision with the strongest ordinary domain-native baseline;
8. record the minimal restorative descent set;
9. fail the translation if success requires post-hoc role invention, semantic substitution, or changing the source contract.

A positive mapping is therefore insufficient by itself. A serious cross-domain object must also possess a **refusal property**: it must be able to say that an apparently attractive correspondence is not present.

8. Prospective conceptual test I: metrology to software change control

The first translation test used two operationally unlike domains. It is a prospective **conceptual** test of the architecture, not an empirical benchmark study.

8.1 Source: correlated measurement packets

Let two calibrated measurement packets estimate the same quantity x :

$$y_i = x + b + e_i$$

with

$$\text{Var}(b) = 4, \quad \text{Var}(e_i) = 1$$

where b is a common calibration bias and e_i is channel-specific noise. Each packet alone has marginal variance

$$\text{Var}(y_i) = 5$$

For a single-channel future-use family, storing an estimate plus marginal variance can be sufficient. For later two-channel fusion, the common-source relation becomes decision-bearing.

If the shared calibration relation is deleted and the packets are treated as independent, naïve fusion reports

$$\left(\frac{1}{5} + \frac{1}{5}\right)^{-1} = 2.5$$

Retaining the covariance

$$\Sigma = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

gives equal-weight fused variance

$$\frac{5 + 5 + 2(4)}{4} = 4.5$$

The omitted shared-source relation therefore changes the uncertainty attached to the upper result. Under the frozen schema, the decision is **REOPEN/REFINE before upper reuse**, with descent targeted at the covariance/calibration relation rather than all raw data. Standard covariance-aware statistical modelling reaches the same local decision [6].

8.2 Target: versioned software release composition

The target was constructed from software/change-control semantics rather than by translating metrology vocabulary.

A certified build **B@1.4** and an independently certified component **P@2.1** are later composed into release **R**. Both load one shared runtime dependency **L**, but:

B@1.4 requires L@2.x under compatibility contract C2

P@2.1 requires L@3.x under compatibility contract C3.

Each constituent can legitimately retain its local PASS state. The upper release is not automatically safe to reuse if the shared dependency relation is hidden. The pre-registered ablation deletes from the upper interface the fact that both components resolve a single **L** and suppresses their version/ABI constraints.

The frozen schema again returns **REOPEN/REFINE before reuse**. The minimal descent set is the two dependency manifests/contracts plus the upper resolution policy/graph and the integration evidence affected by the selected resolution. It does not require reopening unrelated source code or the complete build history.

A conventional dependency resolver, SBOM/provenance system, and integration policy should reach the same local decision [7]. The result is therefore **structural translation with baseline parity**, not software superiority.

8.3 Translation verdict

The structural function remained stable:

- lower objects locally valid
- + shared relation becomes decision-bearing at wider composition
- + relation omitted from upper interface
- upper closure is not future-sufficient
- targeted REOPEN/REFINE restores the missing dependency

Calibration was not relabelled as approval; numeric uncertainty was not forced into software; authority was not used to manufacture symmetry. The target correspondence was at the function of the typed relation, not at the surface word.

This is one positive translation result. It is not evidence of broad domain invariance.

9. Negative control: formal-proof validity and false-symmetry refusal

A second test deliberately used a domain in which two previously instantiated fields are genuinely absent from the target validity relation.

Let L be a formally checked theorem certificate under exact environment V_1 , later reused as a dependency in a larger theorem H . Relevant target fields include:

- theorem/proof identity and scope;
- axiom, definition, imported-lemma, library, and trust/kernel dependencies;
- exact environment fingerprint;
- proof provenance/descent route;
- recursive reuse inside H .

Two fields are **not** constitutive of derivability relative to V_1 :

1. live editorial/maintainer authority or permission;
2. measurement-style statistical uncertainty.

A maintainer may control whether a proof is merged or published, but that workflow permission does not make an invalid derivation valid. Human confidence is not measurement uncertainty, and an open axiom is a typed dependency rather than a variance term.

9.1 Control A: genuine absence

All mathematical dependencies remain unchanged and the authority field is simply absent.

Predicted and obtained schema decision:

PRESERVE.

No reopening is triggered merely because another domain had an authority field.

9.2 Control B: dependency ablation

The top-level theorem name and statement are retained, but the exact environment fingerprint for one load-bearing definition or imported lemma K is hidden while the target environment changes materially.

Predicted and obtained schema decision:

REOPEN/REFINE the affected dependency/certificate chain before reuse.

A proof assistant or content-addressed dependency checker likewise invalidates or rechecks a certificate when its formal environment no longer matches [9].

9.3 Refusal-test verdict

The schema distinguished:

Field absent because it is not constitutive here.

from

Load-bearing relation existed but was omitted from the closure.

That refusal property matters because it reduces the risk that cross-domain success is generated by forcing every field into every target. Again, domain-native proof tooling already solves the local validity problem. The earned result is typed cross-domain transport at baseline parity.

10. Bell/CHSH formalisation: attached semantics, tetrahedral facets, and the new-physics gate

Version 0.3 stopped at a correct negative result: an unrestricted four-correlator object cannot be losslessly encoded by one ordinary three-degree-of-freedom tetrahedral barycentric point. Version 0.4 treats that result as the start of the derivation rather than the end. The question is whether a **lawful tetrahedral object** exists at the correct Bell address, whether the semantic attachment can be stated exactly, and whether any residual survives after established Bell geometry is subtracted.

The answer is split. A tetrahedron is not privileged for the whole unrestricted correlator space. However, the local CHSH correlation polytope has genuine three-dimensional tetrahedral facets inside $\mathbb{R}^4$ [25,26]. This permits a complete **facet-plus-normal** chart of the four-correlator object. The chart is exact but is a reparameterisation; by itself it cannot constitute new physics.

10.1 Native Bell behaviour and correlator objects

For outcomes $a, b \in \{-1, +1\}$ and settings $x, y \in \{0, 1\}$, write the full binary behaviour as

$$P = \{p(a, b \mid x, y)\}.$$

Define context-wise first moments and correlators

$$A_{xy} = \sum_{a,b} a p(a, b \mid x, y), \quad B_{xy} = \sum_{a,b} b p(a, b \mid x, y),$$

$$E_{xy} = \sum_{a,b} ab p(a, b \mid x, y).$$

For each context,

$$p(a, b \mid x, y) = \frac{1}{4} [1 + aA_{xy} + bB_{xy} + abE_{xy}],$$

subject to positivity and normalisation. Under operational no-signalling, $A_{xy} = A_x$ and $B_{xy} = B_y$. The correlator vector is

$$E = (E_{00}, E_{01}, E_{10}, E_{11})^\top \in [-1, 1]^4.$$

The standard CHSH orientation is

$$c = (1, 1, 1, -1)^\top, \quad S_c(E) = c^\top E = E_{00} + E_{01} + E_{10} - E_{11}.$$

The local bound is $S_c \leq 2$ [15,23]. Cirel'son/Tsirelson, Landau, and Masanes remain the native owners of the corresponding quantum correlator constraints [16,18,19]. Fine remains the native owner of the Bell-local/global-joint equivalence at the CHSH scope [15].

10.2 The first tetrahedron: a four-context indexing simplex and its face lattice

Let

$$\mathcal{C} = \{00, 01, 10, 11\}$$

be the four Bell contexts. Treating these four **labels** as vertices of an abstract 3-simplex is an indexing device, not a claim that the physical correlator space is three-dimensional. A correlator object is a vertex field

$$E : \mathcal{C} \rightarrow [-1, 1],$$

and a CHSH sign attachment is a section

$$\sigma : \mathcal{C} \rightarrow \{-1, +1\}, \quad \prod_{c \in \mathcal{C}} \sigma(c) = -1.$$

The odd-parity condition gives the eight CHSH sign orientations. Let

$$c_\sigma = (\sigma(00), \sigma(01), \sigma(10), \sigma(11))^\top,$$

and define

$$S_\sigma(E) = c_\sigma^\top E = \sum_{c \in \mathcal{C}} \sigma(c) E_c.$$

To formalise attachment erasure, let $\pi_{\text{bag}}(\sigma)$ retain only the multiset of four signs and forget which context receives which sign. This gives an exact Bell instance of Section 3.6: the coefficient inventory can stay unchanged while the operational selector changes because its attachment changed.

Proposition 6 - attached-sign facet selection. Let $\sigma \neq \sigma'$ be two CHSH sign sections with $\pi_{\text{bag}}(\sigma) = \pi_{\text{bag}}(\sigma')$; for example, the single negative sign is attached to different contexts. Then there exists a correlator vector E such that $S_\sigma(E) \neq S_{\sigma'}(E)$. Consequently, preserving the unattached sign inventory does not preserve the CHSH functional.

Proof. Choose a context c_0 for which $\sigma(c_0) \neq \sigma'(c_0)$ and let E have $E_{c_0} = 1$ and every other component zero. Then $S_\sigma(E) - S_{\sigma'}(E) = \sigma(c_0) - \sigma'(c_0) = \pm 2$. $\square$

This proposition is deliberately narrow: it establishes functional load-bearingness of Bell coefficient-to-context attachment on a fixed labelled correlator vector. It is not a general theorem about natural-language semantics and does not assign physical causal power to signs.

The ordinary face lattice of the context simplex is the Boolean lattice

$$\mathcal{L}_{\mathcal{C}} = (2^{\mathcal{C}}, \subseteq, \cap, \cup).$$

For $U \subseteq \mathcal{C}$, define the restriction

$$r_U(E, \sigma) = (E|_U, \sigma|_U).$$

Proposition 7 - strict-face insufficiency for unrestricted CHSH evaluation. If $U \subsetneq \mathcal{C}$ and omitted correlators are not guaranteed recoverable, then r_U is not future-sufficient for exact evaluation of S_σ over unrestricted correlator inputs.

Proof. Every CHSH coefficient is non-zero. Choose $c_0 \notin U$ and two correlator vectors that agree on U but differ at c_0 . Their restricted interfaces are identical while their S_σ values differ. $\square$

This is the mathematically sober content of the “lattice” branch at this address: a face/restriction lattice for context-labelled information and a preserve-or-reopen condition when a future operation needs a missing vertex field. It is not a newly postulated physical lattice.

10.3 The second tetrahedron: an established CHSH facet in $\mathbb{R}^4$

The Bell-native local correlator polytope is

$$\mathcal{L}_E = \text{conv} \left\{ E \in \{-1, +1\}^4 : E_{00}E_{01}E_{10}E_{11} = 1 \right\}.$$

It has eight vertices and is full-dimensional in $\mathbb{R}^4$. Its sixteen tetrahedral facets split into eight E-type facets $\pm E_{xy} \leq 1$ and eight CHSH-type facets [25,26]. The E-type inequalities are genuine facets of $\mathcal{L}_E$, not merely ambient-cube bounds: fixing one correlator at ± 1 leaves four affinely independent local vertices on the supporting hyperplane [26]. For the standard orientation $c = (1, 1, 1, -1)^\top$, the CHSH facet is

$$\mathcal{F}_c = \{E \in \mathcal{L}_E : c^\top E = 2\}.$$

One convenient ordered set of its four deterministic local vertices is

$$v_1 = (-1, 1, 1, -1)^\top, \quad v_2 = (1, -1, 1, -1)^\top,$$

$$v_3 = (1, 1, -1, -1)^\top, \quad v_4 = (1, 1, 1, 1)^\top.$$

Each satisfies $c^\top v_i = 2$. Put them into the matrix

$$V = \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ -1 & -1 & -1 & 1 \end{pmatrix}.$$

Direct calculation gives

$$\det V = 16,$$

and

$$V^{-1} = \frac{1}{4} \begin{pmatrix} -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}.$$

All six pairwise distances between the v_i equal $2\sqrt{2}$, so this facet is a regular tetrahedron in its three-dimensional affine hyperplane. Its Euclidean three-volume is

$$\text{Vol}_3(\mathcal{F}_c) = \frac{8}{3}.$$

The crucial distinction is now explicit: the **context indexing simplex** of Section 10.2 has vertices labelled by measurement contexts, whereas the **CHSH facet tetrahedron** has vertices labelled by local deterministic correlator strategies. They both have four vertices, but they are not the same mathematical object and should not be merged by the drawing.

10.4 Facet-adapted Tetrahedral Bell Chart

The v0.3 dimension objection can now be solved exactly rather than evaded. Define the signed transverse CHSH coordinate

$$\boxed{\nu_c(E) = c^\top E - 2 = S_c(E) - 2.}$$

Since $\|c\|^2 = 4$, the orthogonal projection of E onto the affine CHSH hyperplane $c^\top u = 2$ is

$$\boxed{\Pi_c(E) = E - \frac{\nu_c(E)}{4}c.}$$

Define the facet coordinates

$$\boxed{\lambda_c(E) = V^{-1}\Pi_c(E).}$$

Proposition 8 - tetrahedral-plus-normal completeness. The map

$$\Phi_c : E \mapsto (\lambda_c(E), \nu_c(E))$$

is an affine bijection

$$\mathbb{R}^4 \longleftrightarrow \{(\lambda, \nu) \in \mathbb{R}^4 \times \mathbb{R} : \mathbf{1}^\top \lambda = 1\},$$

with inverse

$$\boxed{E = V\lambda + \frac{\nu}{4}c, \quad \mathbf{1}^\top \lambda = 1.}$$

The constrained codomain is four-dimensional despite displaying five scalar entries: λ contributes three independent affine coordinates because $\mathbf{1}^\top \lambda = 1$, and ν contributes the fourth.

Proof. By construction, $c^\top \Pi_c(E) = 2$. Also $c^\top V = 2\mathbf{1}^\top$. Therefore, with $\lambda = V^{-1}\Pi_c(E)$,

$$2 = c^\top V\lambda = 2\mathbf{1}^\top \lambda,$$

so $\mathbf{1}^\top \lambda = 1$. Rearranging the projection definition gives $E = V\lambda + (\nu/4)c$. Conversely, any (λ, ν) with $\mathbf{1}^\top \lambda = 1$ produces a unique E , and applying the definitions recovers the same pair because V is invertible. $\square$

We call Φ_c the **Facet-adapted Tetrahedral Bell Chart (TBC)**. It resolves the degree-of-freedom accounting precisely:

- λ has four components with one affine constraint, hence three independent coordinates;
- ν is the fourth independent coordinate;
- on the local CHSH facet, $\nu = 0$ and $\lambda_i \geq 0$;
- for a general E , λ can leave the non-negative tetrahedron even though the affine chart remains exact.

Thus v0.3 was correct that a tetrahedral barycentric point alone cannot encode unrestricted E . The complete object is **tetrahedral base plus transverse coordinate**, not tetrahedron alone.

For any symmetry-equivalent CHSH orientation c_σ , the same construction is obtained by the corresponding Bell relabelling. More explicitly, let V_σ be the 4×4 matrix whose columns are the four local vertices saturating $c_\sigma^\top E = 2$. Then $|\det V_\sigma| = 16$ and $c_\sigma^\top V_\sigma = 2\mathbf{1}^\top$, so the formulas above carry over with (c, V) replaced by (c_σ, V_σ) . The physics does not depend on privileging the displayed standard orientation.

10.5 Semantic meaning of the chart coordinates

The chart coordinates have deliberately narrow semantics.

1. c_σ / σ : the exact attachment of CHSH coefficients to measurement contexts; changing the attachment selects a different Bell facet functional.
2. λ_σ : barycentric coordinates of the orthogonal projection onto the selected CHSH hyperplane. When $\nu = 0$ and $\lambda_i \geq 0$, they are ordinary convex weights over the four local deterministic vertices of that facet. When $\nu \neq 0$, they remain projection coordinates but are not by themselves a hidden-variable decomposition of E .
3. ν_σ : signed CHSH excess over the local facet, $S_\sigma - 2$. It is not a new field or new observable; it is an existing Bell functional written as the transverse chart coordinate.
4. $\mathcal{L}_C$: the context face lattice, which tracks exposed/restricted context-labelled quantities. It is an information-address structure, not a physical spacetime or causal lattice.

This division prevents the word “tetrahedron” from silently mixing index semantics, local deterministic mixtures, and physical correlator geometry.

10.6 A normal-ray chain: local facet, Tsirelson point, and PR point

The centroid of the standard CHSH facet is

$$E_0 = V \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right)^\top = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2} \right)^\top = \frac{1}{2}c.$$

Move outward along the facet normal:

$$E(r) = E_0 + rc = \left(\frac{1}{2} + r, \frac{1}{2} + r, \frac{1}{2} + r, -\frac{1}{2} - r \right)^\top.$$

Then

$$S_c(E(r)) = 2 + 4r, \quad \nu_c(E(r)) = 4r,$$

while

$$\lambda_c(E(r)) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right)^\top$$

is fixed. Three established benchmark points therefore lie on one exact chart line:

Point	r	ν	S
local-facet centroid	0	0	2
symmetric Tsirelson point $E = (1/\sqrt{2})c$	$(\sqrt{2} - 1)/2$	$2(\sqrt{2} - 1)$	$2\sqrt{2}$
PR-box correlator / algebraic point $E = c$ [24]	1/2	2	4

These values are not new predictions. They show that the chart cleanly separates a three-coordinate facet address from a one-coordinate displacement through local, quantum-nonlocal, and post-quantum no-signalling correlator regimes.

10.7 Pullback of the native quantum boundary

At the unbiased-marginal correlator scope, the Tsirelson-Landau-Masanes (TLM) family gives the native quantum boundary [16,18,19]. A representative orientation is

$$|\arcsin E_{00} + \arcsin E_{01} + \arcsin E_{10} - \arcsin E_{11}| \leq \pi,$$

with the appropriate symmetry-equivalent family required for the complete correlator criterion. In TBC coordinates one performs only the substitution

$$E(\lambda, \nu) = V\lambda + \frac{\nu}{4}c.$$

Define the chart pullback of the quantum correlator set by

$$\mathcal{Q}_c^{\text{TBC}} = \left\{ (\lambda, \nu) : \mathbf{1}^T \lambda = 1, V\lambda + \frac{\nu}{4}c \in \mathcal{Q}_E \right\}.$$

Where the vertical section is non-empty, one may define an implicit quantum ceiling

$$\nu_Q(\lambda) = \sup \left\{ \nu : V\lambda + \frac{\nu}{4}c \in \mathcal{Q}_E \right\}.$$

This is not a newly derived quantum boundary; it is the native boundary expressed in the chart.

At the facet centroid $\lambda_* = (1/4, 1/4, 1/4, 1/4)$, write

$$E = \left(\frac{1}{2} + \frac{\nu}{4} \right) c.$$

The representative TLM condition on the outward ray becomes

$$4 \arcsin \left(\frac{1}{2} + \frac{\nu}{4} \right) \leq \pi,$$

so

$$\boxed{\nu \leq 2(\sqrt{2} - 1)}$$

for the quantum correlator boundary on that ray. The correlator cube permits up to $|E_{xy}| \leq 1$, giving the algebraic/no-signalling correlator ceiling $\nu \leq 2$. The chart therefore reproduces, but does not explain beyond native theory, the familiar ordering

$$0 < 2(\sqrt{2} - 1) < 2$$

for local-facet excess, symmetric quantum maximum, and algebraic maximum.

10.8 Marginal fibre: why tetrahedral correlator coordinates do not close full behaviour

The projection from full behaviours to correlators is

$$\pi_E : P \longmapsto E.$$

For fixed E , the inverse image

$$\pi_E^{-1}(E)$$

is a set-theoretic **fibre** of behaviours sharing the same correlators. No fibre-bundle structure is assumed. Using the binary parametrisation,

$$p(a, b \mid x, y) = \frac{1}{4} \left[1 + aA_{xy} + bB_{xy} + ab \left(V\lambda + \frac{\nu}{4}c \right)_{xy} \right].$$

Thus (λ, ν) reconstructs E exactly but does not reconstruct the marginal coordinates A_{xy}, B_{xy} . Under no-signalling these reduce to A_x, B_y , but they still remain additional information subject to positivity constraints.

The v0.3 same-correlator/different-signalling construction is therefore now a direct fibre statement. Let P_0 have $A = B = E = 0$. Let P_{sig} have

$$A_{xy} = \frac{1}{2}(-1)^y, \quad B_{xy} = 0, \quad E_{xy} = 0.$$

Both have identical E , hence identical TBC coordinates for any fixed CHSH orientation, but Alice's +1 marginal is $3/4$ for $y = 0$ and $1/4$ for $y = 1$ in P_{sig} . Therefore a TBC interface is future-sufficient for declared correlator questions but not for general no-signalling/full-behaviour questions. Widening the operation family forces reopening of the marginal-bearing fibre [17].

10.9 Exact one-correlator ablation in TBC coordinates

Let

$$t = \frac{1}{\sqrt{2}}, \quad E_Q = (t, t, t, -t)^\top.$$

Then

$$\lambda(E_Q) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right)^\top, \quad \nu(E_Q) = 2(\sqrt{2} - 1).$$

Neutralise only E_{11} at the object level:

$$E_A = (t, t, t, 0)^\top.$$

This gives

$$\nu(E_A) = -2 + \frac{3\sqrt{2}}{2},$$

and

$$\lambda(E_A) = \left(\frac{1}{4} - \frac{\sqrt{2}}{16}, \frac{1}{4} - \frac{\sqrt{2}}{16}, \frac{1}{4} - \frac{\sqrt{2}}{16}, \frac{1}{4} + \frac{3\sqrt{2}}{16} \right)^\top.$$

Both E_Q and E_A satisfy the native TLM correlator criterion at the stated unbiased-marginal scope. The ablation changes both the transverse coordinate and the projected tetrahedral address. Restoring $E_{11} = -t$ restores the original pair (λ, ν) exactly. This is a surgical representation-level load-bearing test; it is not evidence that an experimental intervention can vary one correlator independently while holding the underlying physical realisation fixed.

10.10 Hypervolume candidate and exact collapse to CHSH excess

A natural attempt to make the tetrahedron carry additional physics is to use the four-dimensional volume generated by the facet and an off-facet point. Define

$$\mathcal{V}_c(E) = \text{Vol}_4(\text{conv}\{v_1, v_2, v_3, v_4, E\}).$$

The perpendicular distance from E to the CHSH hyperplane is

$$h(E) = \frac{|c^\top E - 2|}{\|c\|} = \frac{|\nu|}{2}.$$

Since the facet tetrahedron has three-volume $8/3$, the four-dimensional pyramid formula gives

$$\mathcal{V}_c(E) = \frac{1}{4} \left(\frac{8}{3} \right) \left(\frac{|\nu|}{2} \right).$$

The volume used here is Euclidean volume in the standard correlator embedding $\mathbb{R}^4$; no coordinate-free physical volume is being asserted.

Therefore:

Proposition 9 - hypervolume collapse. For the standard CHSH facet,

$$\boxed{\mathcal{V}_c(E) = \frac{|\nu_c(E)|}{3} = \frac{|S_c(E) - 2|}{3}}.$$

The same result holds for symmetry-equivalent CHSH facets under the corresponding orthogonal relabellings.

This is a useful negative result. The most immediate tetrahedral four-volume is not an independent physical invariant; it is exactly a rescaling of the known CHSH excess. The geometry does real bookkeeping work, but it does not generate a new Bell quantity.

10.11 Reparameterisation null and the new-physics gate

The TBC is invertible. That fact creates a hard scientific boundary.

Proposition 10 - reparameterisation null. Let Φ be a bijection between two coordinate descriptions of the same physical behaviour, and let every prediction, admissibility test, and experimental statistic in the second description be the pullback or pushforward of the first through Φ . Then no experiment can distinguish the descriptions solely by their coordinate choice.

Proof. Every physical input has one corresponding coordinate image and every declared statistic has, by assumption, the same numerical value after composition with Φ or Φ^{-1} . Therefore the induced probability distributions over experimental outcomes are identical. $\square$

Consequently, the TBC itself cannot be new physics. A genuine physical extension would require an additional law, constraint, or dynamics that is **not** merely a rewritten native Bell/QM relation. Schematically, one would need something of the form

$$G_{\text{new}}(\lambda, \nu, m; \theta) = 0$$

or

$$p_{\text{new}}(a, b \mid x, y; \theta) = p_{\text{QM}}(a, b \mid x, y) + \delta p_\theta(a, b \mid x, y),$$

with all new variables and couplings typed, positivity and normalisation preserved, the intended signalling/causal constraints stated, a standard/null limit such as $\delta p_{\theta_0} = 0$, and at least one prospectively measurable discriminator. The present derivation supplies no non-zero δp_θ and no independent G_{new} after native-owner subtraction.

10.12 Full Bell/tetrahedral fixed-point verdict

The reopened Bell branch now returns the following fixed point:

Question	v0.4 result
Does an unrestricted four-correlator vector fit in one ordinary tetrahedral barycentric point?	No
Is there a genuine tetrahedral Bell object at a narrower native address?	Yes - each CHSH facet is a 3D tetrahedron in $\mathbb{R}^4$
Can unrestricted $E \in \mathbb{R}^4$ be reconstructed from a tetrahedral base plus one extra coordinate?	Yes - exactly, via $E = V\lambda + (\nu/4)c$
Is the CHSH sign/context semantics mathematically attachable?	Yes - $\sigma : \mathcal{C} \rightarrow \{\pm 1\}$ selects S_σ and the facet orientation
Does the context “lattice” have a sober mathematical owner?	Yes - the Boolean face/restriction lattice $2^{\mathcal{C}}$
Does TBC alone encode full Bell behaviour?	No - marginals remain in the fibre of $P \mapsto E$
Does the chart reproduce local/Tsirelson/PR ordering on the symmetric normal ray?	Yes
Does the natural 4-volume give a new invariant?	No - it collapses to $S - 2 /3$
Does TBC derive a new quantum boundary?	No - TLM/Tsirelson are only pulled back into new coordinates
Does any independent new-physics residual survive?	No - current physical residual NULL

A tetrahedral representation therefore survives at this Bell-native facet address, but in a narrower and more rigorous form than the originating sketch: **a facet-adapted coordinate and semantic-address construction on established Bell geometry**. This does not establish that the originating cross-domain relational tetrahedron is universally realised by Bell geometry. What dies is the stronger claim that tetrahedral shape itself supplies a new law or physical mechanism.

11. What closed and what survived

The cumulative pressure-testing is easiest to read as explicit contractions and surviving objects.

Candidate claim	Current status
A literal new physical dimension follows from the originating drawing	Closed / unsupported
Three lower roles or tetrahedral arity are universally privileged	Closed / unsupported
A tetrahedral barycentric point alone losslessly represents unrestricted CHSH correlators	Rejected
A genuine tetrahedral object occurs in CHSH-native geometry	Yes - the 3D CHSH facets of the 4D local correlator polytope
Tetrahedral base plus transverse coordinate exactly reconstructs unrestricted $E \in \mathbb{R}^4$	Proved as an affine chart
CHSH coefficient-to-context attachment can be decision-bearing	Proved for distinct sign sections with the same erased sign multiset
The context lattice is a new physical lattice	No - ordinary Boolean face/restriction lattice only
The TBC coordinates determine arbitrary full Bell behaviour	No - marginal fibre must be reopened
Natural facet-to-point 4-volume is a new Bell invariant	No - exactly $S - 2 /3$
TBC independently derives TLM/Tsirelson	No - coordinate pullback only
TBC produces a new Bell inequality or mechanism	No - physical residual NULL
Parent completion, by itself, is new mathematics	No - native satisfiability/global-consistency machinery owns this address

Candidate claim	Current status
Exact relation ablation can be an operational three-valued test	Supported as a framework definition
Future sufficiency is operation-relative and can fail under widened reuse	Supported formally and constructively
Strong domain-native baselines reproduce the local metrology/software/proof decisions	Supported: baseline parity
Broad cross-domain invariance or practical superiority	Open; requires held-out evaluation

The strongest current statement is therefore sharper than v0.3 but not physically stronger than the mathematics earns:

The paper now supplies an exact operator-attachment semantics and a complete facet-adapted tetrahedral-plus-normal chart for CHSH correlators, while the full native-owner squeeze shows that the chart is a representation of known Bell structure rather than a new physical law.

That result is still load-bearing for the cross-domain schema: it demonstrates how an evocative geometric candidate can be carried from sketch to exact mathematics, split into distinct typed objects, tested for loss, and either retained as a lawful chart or killed as an unsupported mechanism.

12. Relation to ordinary formal machinery

The current result is intentionally tested against ordinary machinery rather than insulated from it.

12.1 Sufficient statistics and task-relative compression

Classical sufficiency asks whether a statistic retains the information required for a target inference [1]. Information-bottleneck methods explicitly optimize compression relative to relevant information [2]. The present future-sufficiency rule is therefore not a new compression theory. Its added engineering emphasis is that an interface may remain deliberately incomplete if it also carries a guaranteed restorative-descent route before a widened task uses omitted information.

12.2 Factor and dependency graphs

Factor graphs expose how global functions depend on local factors [3]. Dependency DAGs and provenance systems likewise make hidden shared sources and transitive impact cones explicit. The software and metrology tests are therefore not evidence that the schema beats graph methods; the graph methods are strong baselines and frequently the correct local implementation.

12.3 Sheaf-like local-to-global consistency

Sheaves provide a principled language for heterogeneous local information, overlap consistency, and global integration [4]. This is close to the schema’s demand that local interfaces agree where they claim to share a dependency while retaining domain-specific information elsewhere.

12.4 Open-system composition

Compositional formalisms can “black-box” internal structure while preserving externally relevant behaviour and compose systems through typed interfaces [5]. This is a particularly close baseline for the whole-downward/part-upward role change of a reusable closure.

12.5 Provenance, access control, and proof checking

Software supply-chain provenance demonstrates that closed artifacts can remain valid only relative to traceable build/dependency history [7]. Access-control models demonstrate that permission is a typed relation, not merely knowledge that a user or role exists [8]. Proof assistants demonstrate a clean case where mathematical validity depends on an explicit checking environment rather than workflow authority [9].

These comparisons are not weaknesses of the paper. They define its proper burden. A cross-domain schema is scientifically useful only if it can integrate these distinctions without pretending to replace the mature local machinery that already handles them well.

12.6 Addressed Admissible Futures as direct first-party precedent

The earlier *Addressed Admissible Futures: Future-Sufficient State, Load-Bearing Relations, and Restorative Reachability* already defines future-equivalence/address sufficiency, restorative futures and reserve, counterfactual load-bearing relations under removal/deformation, and recursive readdressing [11]. The present paper therefore does not claim first invention of **load-bearing relation**, ‘future sufficiency,’ restorative reachability, or recursion. Its narrower residual is the cross-domain **interface-composition** problem: how those ideas behave when a locally closed object is reused as a constituent of a wider object, with typed partial instantiation, exact relation-address ablation, refusal, parent-completion checks, and minimum lawful descent.

12.7 View/query determinacy

Database view determinacy is a particularly close comparator. A set of views determines a query when the view information is sufficient to uniquely determine the query answer [12]. That is structurally close to target-relative future sufficiency: two lower states that collapse to the same interface must not split the declared operation result. The present paper does not claim to rediscover determinacy. Its additional interface concern is what happens when the operation family later widens and the system must either preserve extra information or carry a prospective restoration route to the omitted lower relation.

12.8 Counterexample-guided abstraction refinement

Counterexample-guided abstraction refinement (CEGAR) begins from a coarse abstraction and selectively adds precision when the abstraction is too coarse for the property being checked [13]. This strongly constrains any novelty claim for the generic idea “refine when abstraction becomes insufficient.” The present schema’s residual burden is narrower: exact typed relation addresses, partially instantiated fields, a refusal property, provenance-aware descent, and recursive reuse across heterogeneous domains. If those additions do not improve localisation, auditability, or transfer, the architecture should contract toward ordinary abstraction-refinement language.

12.9 Bell typed-boundary prior work

The author’s earlier *Bell Constraints as Typed Boundaries* preprint records the direct first-party Bell provenance and, importantly, already closes the claim that the typed-boundary architecture independently generated a new Bell inequality or Tsirelson derivation [22]. The present Bell section therefore does not reopen that null. Version 0.4 reopens only the representation branch: it formalises the CHSH-facet chart while preserving the earlier null on independent Bell physics and keeping native Bell mathematics with its established owners.

12.10 Global-section and contextuality comparators

Sheaf-theoretic contextuality expresses non-locality/contextuality as obstruction to global sections over compatible local data [14]. That is a strong native comparator for any local-to-global or parent-completion language in the Bell section. It prevents the generic parent-completion wrapper from being mistaken for a new contextuality theorem. The paper uses this comparator affirmatively: the more exact the native owner, the narrower and more testable the residual cross-domain interface claim becomes.

12.11 Bell correlation-polytope and tetrahedral-facet geometry

The tetrahedral facet itself is established Bell geometry rather than a new mathematical discovery of this paper. Scarani’s analytical treatment places the CHSH local correlator polytope in $\mathbb{R}^4$, identifies its local correlator vertices, and derives the facet inequalities [25]. Li states the sharper finite inventory used here: the $(2, 2)$ local correlation polytope is a full-dimensional four-dimensional cross-polytope with eight vertices and sixteen tetrahedral facets, split into eight E-type facets and eight CHSH-type facets, four vertices per facet [26]. Gigena, Scala, and Mandarino provide a constructive geometric treatment of the local CHSH set and its facets [27].

Accordingly, no novelty claim is made for “a tetrahedron exists in CHSH.” The paper-specific construction is the typed integration of (i) exact sign-to-context attachment, (ii) the context face/restriction lattice, (iii) a facet-adapted affine chart with one transverse coordinate, (iv) the marginal-fibre reopening boundary, and (v) explicit collapse tests showing when geometric quantities reduce to native Bell functionals. Even that integrated construction is presented as a formalisation and audit object, not as a claimed historically first theorem.

13. Practical significance: the next burden

Baseline parity changes the research question. The next test is not whether the schema can reproduce an obvious decision that local specialists already know how to make. It is whether a common contract can reduce failure at **interfaces between domains, teams, abstractions, or recursive levels**.

Useful measures include:

- omission/error detection rate;
- frequency of hidden shared-dependency loss;
- correctness and size of the selected restorative descent set;
- reconstruction cost after an upper closure fails;
- review time and review burden;
- transfer time when an object moves between teams or domains;
- ambiguity or disagreement about what a “closed” object is allowed to support;
- false-positive reopening caused by requiring fields that are genuinely absent;
- false-negative preservation when a relation was decision-bearing but omitted.

The practical hypothesis is:

A frozen cross-domain contract can reduce interface loss by forcing target, type, dependency, remainder, provenance, authority where applicable, and reopen conditions to be checked together before recursive reuse.

That hypothesis remains unproven. It should be evaluated against strong local checklists, dependency/provenance systems, view-determinacy and abstraction-refinement baselines, review workflows, and domain-specific interface specifications. If the common schema adds no measurable improvement and only redescribes what each local baseline already does at equal or lower burden, its practical significance contracts to common-language convenience.

The Bell continuation now begins from a stronger fixed object. The TBC mapping, semantic attachment, context face lattice, marginal-fibre boundary, and reparameterisation-null rule can all be frozen before any new scenario is inspected. A later held-out phase should compare preserve/refine/reopen decisions and representation burden against strong Bell-native baselines, then transport the same rules to a non-CHSH Bell scenario where four-context/tetrahedral convenience is absent. That is the cleanest test of whether the interface method travels or merely overfits CHSH geometry. A physical claim remains gated by a non-reparameterisation residual with a standard/null limit and an observable discriminator; v0.4 still supplies none.

14. Falsifiers and contraction rules

The architecture should be weakened, reduced, or rejected when any of the following occurs:

1. **Post-hoc semantic repair:** a target translation succeeds only after roles are redefined in response to the observed outcome.
2. **Forced field symmetry:** the schema invents authority, uncertainty, observer state, or another field merely because a source domain had it.
3. **Anything-goes optionality:** fields can be removed without a typed reason, so every failure can be re-described as lawful absence.
4. **Dependency blindness:** deletion of a known decision-bearing relation does not trigger refinement or reopening.
5. **No refusal property:** every target can be made to fit by renaming generic concepts such as “dependency” or “context”.

6. **No minimal descent:** the schema can only respond to strain by reopening the entire history, defeating the compression/interface purpose.
7. **Cardinality dependence without evidence:** the formal rule requires three inputs where arbitrary independently justified arity works equally well.
8. **Authority flattening:** descriptive representation of a permission or delegation is treated as enactment of that authority.
9. **Baseline dominance:** an ordinary domain-native or general compositional formalism captures the integrated contract with equal or lower burden and no loss of cross-domain audit function.
10. **No practical gain:** held-out use shows no reduction in interface error, review/transfer burden, ambiguity, or restoration cost relative to strong baselines.
11. **Unbounded generalisation:** one positive translation and one refusal test are reported as universal cross-domain invariance.
12. **Physical overreach:** representational usefulness is converted into an unsupported claim about the geometry or ontology of physical reality.
13. **Ablation ambiguity:** the paper reports a determinate load-bearing verdict where no lawful exact ablation was defined, or where equally lawful ablations disagree.
14. **Restoration-signature blindness:** two RESTORE-REQUIRED states with materially different minimal descent families are treated as equivalent.
15. **Target leakage:** Bell- or domain-specific structure is imported into the generic rule only after the target outcome is inspected.
16. **Native-owner theft:** satisfiability, TLM/Tsirelson, Fine/global-joint completion, view determinacy, CEGAR refinement, or another established result is presented as generated by the generic wrapper.
17. **Bell overreach:** the Phase-0 calibration or the v0.4 TBC formalisation is described as a new Bell equation, quantum boundary, or physical mechanism despite returning a null physical residual.
18. **Tetrahedron conflation:** context labels, local deterministic facet vertices, and physical state coordinates are treated as the same tetrahedral object merely because each set has four elements.
19. **Lost transverse coordinate:** the paper uses λ alone as a lossless representation of unrestricted E while omitting ν .
20. **Barycentric overreach:** λ is interpreted as a probability mixture when $\nu \neq 0$ or when λ has negative components.
21. **Fibre blindness:** correlator/TBC coordinates are claimed sufficient for full-behaviour or no-signalling questions without carrying/reopening marginal information.
22. **Geometric novelty inflation:** ν , the TBC quantum roof, or the four-volume is presented as a new Bell quantity although it is respectively CHSH excess, a pullback of native quantum constraints, or $|S - 2|/3$.
23. **Reparameterisation masquerade:** a coordinate change is presented as new physics without a new law or measurable residual.
24. **Semantic de-attachment:** sign/operator content is transported without its exact context/receiver and then treated as preserved.

A negative result remains informative: it identifies which part of the proposed common contract fails to earn its complexity.

15. Limitations

First, the cross-domain translation tests remain conceptual and internally developed; they are not blinded external experiments or independent replications.

Second, the common schema is broad enough that weak examples could be made to look successful. Freezing, refusal, exact-ablation, and target-leakage controls reduce this risk but do not replace held-out evaluation.

Third, future sufficiency is operation-relative. No interface is claimed sufficient for every future use.

Fourth, minimal restorative descent may be computationally difficult in large cyclic or distributed systems. The paper defines the target object but supplies no new universal algorithm for finding it.

Fifth, exact relation ablation is not always physically or logically available. The explicit unknown state must be retained rather than replaced by an invented *ceteris-paribus* intervention.

Sixth, the semantic attachment tuple $\beta(q)$ is a formal address device. It does not establish that all natural-language meaning can be reduced to one fixed tuple, nor that semantic relations exert new physical causes.

Seventh, the context simplex in Section 10.2 is an indexing construction. Its Boolean face lattice does not imply a physical tetrahedral lattice.

Eighth, the CHSH facet tetrahedron in Section 10.3 is Bell-native geometry already implicit in the local correlation polytope. The paper does not claim first discovery of that facet structure.

Ninth, the TBC is an exact affine reparameterisation of the four-correlator object relative to a selected CHSH facet. Different CHSH orientations give symmetry-equivalent charts. No orientation is physically privileged by this paper.

Tenth, λ is barycentric only for the projected hyperplane point and represents a convex mixture of the four facet vertices only when its entries are non-negative. Away from the facet it must not be read as a hidden-variable probability distribution for the original E .

Eleventh, the TBC is correlator-level. Full behaviours retain marginal degrees of freedom in the fibre of $P \mapsto E$. The chart therefore does not solve arbitrary biased-marginal or signalling questions by itself.

Twelfth, the quantum ceiling $\nu_Q(\lambda)$ is an implicit pullback of established quantum-correlator constraints, not a new derivation of them. The explicit centroid value uses the known TLM/Tsirelson boundary.

Thirteenth, the hypervolume result is deliberately negative for novelty: it collapses exactly to CHSH excess and so cannot be counted as an independent physical invariant.

Fourteenth, no non-zero δp , new coupling, new dynamics, new Bell inequality, or experimentally distinguishable deviation from standard quantum predictions is derived. The present new-physics residual is null.

16. Conclusion

The originating geometric question has now been carried through two different kinds of pressure. The first, completed in v0.3, showed that one ordinary tetrahedral barycentric point cannot losslessly encode an unrestricted four-correlator CHSH object. The second, completed here, asks whether that failure hides a better-typed tetrahedral construction rather than proving that every tetrahedral role is decorative.

The answer is yes, with a strict boundary. The four Bell contexts support an abstract indexing simplex and a Boolean face lattice for exposed versus omitted context-labelled information. Exact sign-to-context attachment defines the CHSH orientation. Separately, Bell-native local correlator geometry supplies actual tetrahedral CHSH facets in $\mathbb{R}^4$. Projecting an arbitrary correlator vector to a selected facet hyperplane and retaining one transverse coordinate yields the exact Facet-adapted Tetrahedral Bell Chart

$$E = V\lambda + \frac{\nu}{4}c, \quad \mathbf{1}^\top \lambda = 1.$$

That equation resolves the four-versus-three count without forcing the tetrahedron to carry a fourth degree of freedom it does not possess. The tetrahedron carries the three affine coordinates of the facet; $\nu = S - 2$ carries the transverse coordinate. The distinction is explicit and reversible.

The chart also clarifies the semantic object. A CHSH sign is not merely one of four visible coefficients; its attachment to an exact measurement context selects the functional and hence the facet orientation. Erasing that attachment can collapse different operational objects. This is the Bell instance of the paper's more general operator-to-target address rule.

The strongest scientific pressure then contracts the geometry. On the facet-centroid normal ray, the chart places the local boundary, symmetric Tsirelson point, and PR point at exact ν values, but those values are

established Bell results. Pulling TLM into the chart gives no new quantum boundary. Full behaviours still require the marginal fibre. The most obvious geometric candidate for a new invariant - the four-volume between the facet and an off-facet point - is exactly $|S - 2|/3$. Finally, the chart is bijective, so a coordinate change alone cannot alter experimental predictions.

The resulting Bell verdict is therefore neither `tetrahedron was wrong` nor `nortetrahedron gives new physics`. The precise result is:

Tetrahedral structure is mathematically legitimate at the CHSH-facet address and can be completed to an exact four-coordinate Bell chart by one transverse coordinate; after native-owner subtraction, the chart remains a representation of known Bell geometry and yields no independent new physical residual.

That is the fixed point supported by the present derivation. A later new-physics branch is allowed only if an additional typed law or dynamics survives reparameterisation, native Bell/QM comparison, standard-limit recovery, and a prospective observable discriminator. No such residual is manufactured here.

The cross-domain compositional claim remains architectural and testable. Version 0.4 strengthens it by demonstrating the full lifecycle the schema is intended to enforce: preserve an initially useful geometric intuition, identify the dimension failure, reopen the exact missing address, distinguish two superficially similar tetrahedra, derive the complete representation, ablate plausible novelty, and retain only the part that continues to do mathematical work.

Declarations

Data and code availability

No new empirical data were generated for this conceptual architecture paper. The reported cases are explicit constructive tests. A companion symbolic verification script for version 0.4 replays the determinant and inverse of the CHSH facet matrix, regular-tetrahedron volume, exact Facet-adapted Tetrahedral Bell Chart inversion, normal-ray benchmark coordinates, the one-correlator ablation coordinates, the hypervolume-collapse identity, and numerical checks of the symmetry-equivalent TLM arcsine family for the two worked correlator points. The script is a reproducibility aid for the displayed algebra, not an empirical validation or a derivation of new Bell physics. Any future empirical benchmark or blinded evaluation should be released with the corresponding protocol and baseline definitions where licensing and ethics permit.

Competing interests

The author declares no competing interests.

Author contribution

Mark Charles McLaughlin: conceptualisation, architecture, methodology, formal analysis, investigation, manuscript direction, review, and accountability for the scientific claims.

AI-assisted drafting and document preparation

OpenAI ChatGPT was used as an AI-assisted drafting, adversarial review, and document-preparation tool during development of the manuscript. The author directed the research object, selected and approved the claims, and retains responsibility for the content.

Version and provenance

Version 0.4 is the substantive next version of the DOI-bearing v0.3 *Cross-Domain Compositional Schema* (10.5281/zenodo.22166005). Version 0.4 carries DOI 10.5281/zenodo.22166468 within the same Zenodo version lineage. It retains the v0.3 typed evaluation/restoration signature, partial exact-relation ablation, structural lemmas, parent-completion boundary, comparator firewall, and Bell Phase-0 calibration, then reopens the previously unresolved tetrahedral Bell branch. The added formalisation includes operator-to-target semantic attachment, the four-context indexing simplex and face lattice, the Bell-native CHSH facet tetrahedron, the exact Facet-adapted Tetrahedral Bell Chart, marginal-fibre reopening, the local/Tsirelson/PR normal-ray calculation, the hypervolume-collapse proposition, and the reparameterisation new-physics gate. The v0.3 and v0.2 DOI objects remain immutable historical versions. The broader relational-architecture

provenance is separately documented in *MKUFT - Layer Before Law* [10], but it is not a framework dependency of the standalone derivations in this paper.

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Appendix A. Minimal notation and decision classes

Symbol	Meaning
X	fuller lower relational object
F	declared future-operation family
C_F	compression/closure operator relative to F
$Y = C_F(X)$	higher-level interface/closure
N_i	typed lower object/node
R_j	typed lower relation/dependency
τ	type/address/scope/source-receiver assignment
α	partially instantiated domain-real auxiliary coordinates
$\text{Eval}_k = (s_k, v_k, \rho_k)$	typed evaluation state, value, and restoration
Abl_j	signature partial protocol-relative exact ablation of relation R_j
$LB_F(R_j; X)$	three-valued load-bearing predicate 0, 1, ?
$\text{Preserve}_k(X, Y)$	direct evaluation-preservation predicate for operation f_k
$FS_F(Y \mid X)$	future-sufficiency predicate
$\text{Comp}_F(X)$	parent-completion predicate
Δ	restorative descent set
P	full Bell conditional behaviour $\{p(a, b \mid x, y)\}$
E_{xy}	Bell correlator
S	CHSH score
$\beta(q)$	typed operator-to-target semantic attachment
$Q = (q, \beta(q))$	addressed operator/modifier object
$\mathcal{C}$	Bell context set $\{00, 01, 10, 11\}$
σ	CHSH sign-to-context attachment section
π_{bag}	erasure retaining the sign multiset while dropping context attachment
c_σ	CHSH coefficient vector induced by σ
$\mathcal{L}_C$	Boolean face/restriction lattice of Bell contexts
$\mathcal{L}_E$	local Bell correlator polytope

Symbol	Meaning
$\mathcal{F}_c$	selected CHSH tetrahedral facet
V	matrix whose columns are the four vertices of $\mathcal{F}_c$
$\nu_c(E)$	signed CHSH excess $c^\top E - 2$
$\Pi_c(E)$	orthogonal projection to the selected CHSH hyperplane
$\lambda_c(E)$	facet barycentric/projection coordinate
Φ_c	Facet-adapted Tetrahedral Bell Chart
π_E	projection from full Bell behaviour to correlator vector
$\mathcal{V}_c(E)$	4-volume of the CHSH facet with apex E
Σ	covariance matrix in the metrology worked test
PRESERVE	current interface remains sufficient for declared use
REJECT	proposed composition is invalid or incompatible as stated
REFINE	interface must expose additional typed state/relation
REOPEN	descend to specified lower structure before reuse
WITHHOLD	information may be sufficient but action is not authorised/admissible